

EXP2-4 – Blamestorming

The **aim** of EXP2-4 is to consider how to plan a set of experiments to investigate a complex problem in materials processing.

The **objectives** for Part 1 of this exercise are:

- Plan a set of experiments to investigate a processing problem
- Work together in your groups
- Write a short experiment plan

We all know what *Brainstorming* is – and we have practiced it in several modules in this degree.

‘Blamestorming’ is a made-up word from the ‘Dilbert’ cartoon series by Scott Adams. It is defined as:

“Sitting around in a group discussing why a deadline was missed or a project failed and who was responsible.”

Background information:

In this exercise we will try to investigate a materials processing problem associated with the manufacture of earthquake bearings. In parts of the world where earthquakes are a risk, such as Los Angeles or Tokyo, all buildings over 3 storeys high are stood on earthquake bearings. These consist of sheets of steel, laminated with sheets of rubber in between. The idea is that the bearing is very stiff in the vertical direction but relatively easy to shear in the horizontal direction. A seismic wave is a transverse elastic wave passing through the earth’s surface. During an earthquake, a building on earthquake bearings remains still while the ground moves back and forth underneath. Rubber is compounded by shear mixing natural or synthetic rubber with sulphur (that creates crosslinks between the elastomer chains) and an initiator or accelerator (a catalyst that starts the cross-linking reaction). The compounded rubber is crosslinked by raising the temperature to obtain the desired elastomeric properties. The rubber is moulded to shape between the steel sheets and heated to form an earthquake bearing with the desired shear stiffness. There may be more than 20 laminations in a large earthquake bearing.

What you should do:

Consider the **‘blamestorming conversation’** written out below. This is all of the information that you will need. In your group, you should **plan an investigation to find out where the problem lies** (not to find out why it has occurred or to rectify the problem). You can assume that all of the materials are still available (from archive stores in the quality system) and that simple equipment is available that would normally be used to test the product – such as mechanical testing equipment and thermal analysis. Advanced analytical equipment or techniques (like electron microscopes and chemical analysis techniques) are unlikely to be available in such a small company and would require the use of a Materials Consultancy or University department at significantly higher cost and longer turn-round time.

The group will present their plan in a short report to the Company Directors, which should **address the following 4 points**:

1. Which factors the group consider to be important that might be causing the material to have changed its properties;
2. What experiments are necessary to identify the factors associated with the changes;
3. An experimental plan for those experiments;
4. An estimate of the time, number of samples and resources, including manpower, required to perform the experiments.

You do not need to explain what an earthquake bearing is or how it is made, the Quakeproof Bearings Company manufacture them, the Company Directors already know this! *Focus on identifying where the problem might come from – not how to fix the problem.*

The report will be **no more than 1000 words excluding author names and references.**

The **title** of the report will be: **EXP2-4 Experimental Plan XX** (where XX is your group number)

The names of **all the group members** should be given as authors of the report.

You will **submit** your group's report as a **pdf document on QMPlus** using the file name:

'QXU5017 EXP2-4 Part 1 Plan XX' (where XX is your group number).

The report must be completed and **submitted by 24:00 on Wednesday 14th May 2025.**

The report must be the **work of the whole group** and the content agreed by all group members.

All group members must hit the submit button to submit the report.

Marks breakdown:

EXP2-4 is 25% of the marks for QXU5017

Part 1 Group Report 1 will be worth 40% of EXP2-4

Part 2 Group Report 2 (Instructions will be given in Week 13) will be worth 60% of EXP2-4

You will conduct a peer review of your group members' contributions to the exercise.

Schedule:

M Groups

Date:	Time:	Place:	Activity:
Wednesday 7 th	10:30-12:10	A310	Introduction to Part 1 of EXP2-4
Friday 9 th	08:30-10:10		Formulate experimental plan
Monday 12 th	10:30-12:10	A310	Complete experimental plan
Wednesday 14 th	10:30-12:10	QM+	Submit Experimental plan
Friday 16 th	08:30-10:10	A310	Introduction to Part 2 of EXP2-4
Monday 19 th	10:30-12:10	A310	Data processing & analysis
Wednesday 21 st	10:30-12:10		Blamestorming
Friday 23 rd	08:30-10:10		Complete and Submit Report

P Groups

Date:	Time:	Place:	Activity:
Wednesday 7 th	08:30-10:10	A310	Introduction to Part 1 of EXP2-4
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Wednesday 21 st	08:30-10:10		Blamestorming
Friday 23 rd	10:30-12:10		Complete and Submit Report

Blamestorming conversation:

It is Monday afternoon at The Quakeproof Bearings Company and an urgent meeting has been called. The conversation goes like this:

Paul (Managing Director): “Sorry for the short notice, but I have called you all together this afternoon because we’ve got a bit of problem. The contractor on the Pacific Towers project in L.A. was on the phone last night. They have had to stop the construction at the 5th floor level because one corner of the building has sunk down and cracks are appearing in the building’s structure. They think there is something wrong with our bearings. The serial numbers of the bearings on that part of the building are 03-714 to 03-815. Any ideas what could cause them to be out of specification?”

Sally (Quality Manager) “hmmm.... those bearings were all made between the middle of July and the end of August.”

Bill (Moulding supervisor): That was while I was off work after my heart by-pass operation. I put young Tony in charge of the moulding operation; I thought he knew what he was doing. You have to get the mould temperature right, you see.”

Tony (Moulding technician): “I followed your instructions exactly. 155°C on the controller.”

Jimmy (Design engineer): “Those bearings were particularly large, maybe they weren’t achieving the moulding temperature all the way through. How long did you leave them in the mould for?”

Tony: “20 minutes, like it says in the procedure thingy.”

Bill: “Maybe that wasn’t long enough for those big ones or the temperature needed to be higher, another 15° C usually does the trick.”

Bindha (Mixing lab): “Wasn’t that about the time we changed the accelerator? Our old supplier closed down and we had to find a new one. It was supposed to be the same type of accelerator but we never had any chemical analysis done.”

Mohammed (Compounding mill operator): “We had that summer internship student, Sophie from Xi’an Jiaotong – Liverpool University, here at that time. I was never convinced she knew what she was doing. The calculated amounts of the sulphur cross-linker are done in volume per cent but we measure out 2 parts per hundred rubber in the compounding mill, what if she got it the wrong way round?”

Bill: “Don’t know what the fuss is about, the Tower of Pisa in Italy wouldn’t be so famous if it didn’t lean over a bit.”

Paul: “Look, this is no laughing matter, we’ve got to get to the bottom of this, and fast. If the contractors sue us for supplying defective bearings and holding up the construction, we’ll be out of business. I want you to tell me how you are going to work out what’s gone wrong with our processing. Come up with a plan to investigate the problem and present it in a short report to the Company Directors by the end of the week. Now jump to it – time is money”

(The characters in this scenario are fictitious; any resemblance to real people or companies is coincidental.)